THE HOM FUNCTOR IS AN FPQC SHEAF

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-hom (canonical tag 08JT). cited at statement line 438. Standing situation.
- Lemma spaces-topologies-lemma-fpqc (canonical tag 03MR). cited at proof line 448. {X_i -> X_T} is an fpqc covering.
- Proposition spaces-descent-proposition-fpqc-descent-quasi-coherent (canonical tag 04W8). cited at proof line 454. Descent of the map u.

Source reference. Original definition 08JT: sources/stacks-chow/tags.txt, source lines 10861–10861. Cited at statement line: the lemma is phrased in this situation; load-bearing.

Source reference. Original lemma 03MR: sources/stacks-chow/tags.txt, source lines 4759–4759. Cited at proof line 448: {X_i -> X_T} is an fpqc covering.

Source reference. Original proposition 04W8: sources/stacks-chow/tags.txt, source lines 6351–6351. Cited at proof line: the descent of the glued morphism; load-bearing.

Lemma 0.2. *In Situation 08JT the functor $Hom(\mathcal{F}, \mathcal{G})$ satisfies the sheaf property for the fpqc topology.*

Proof. Let $\{T_i \to T\}_{i \in I}$ be an fpqc covering of schemes over B. Set $X_i = X_{T_i} = X \times_S T_i$ and $\mathcal{F}_i = u_{T_i}$ and $\mathcal{G}_i = \mathcal{G}_{T_i}$. Note that $\{X_i \to X_T\}_{i \in I}$ is an fpqc covering of X_T, see Topologies on Spaces, Lemma 03MR. Thus a family of maps $u_i : \mathcal{F}_i \to \mathcal{G}_i$ such that u_i and u_j restrict to the same map on $X_{T_i \times_T T_j}$ comes from a unique map $u : \mathcal{F}_T \to \mathcal{G}_T$ by descent (Descent on Spaces, Proposition 04W8). □

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